GG-T-353D
July 14, 1975
SUPERSEDING
Fed. Spec. GG-T-353C
July 15, 1969

FEDERAL SPECIFICATION

THERMOMETERS, SELF-INDICATING, BIMETALLIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers self-indicating bimetallic thermometers for indicating temperatures during the cooking process (see 6.1).

1.2 Classification. Thermometers shall be of the following types, as specified (see 6.2).

Type I - Cooking (roast meat).
Type II - Combination confectioners (candy and cooking, deep fry).
Type III - Oven.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup
PPP-C-843 - Cushioning Material, Cellulosic
PPP-P-291 - Paperboard, Wrapping and Cushioning

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

FSC 6685

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Standard product. The thermometers covered herein shall be a current standard product of the manufacturer, except for any changes necessary to meet specification requirements.

3.2 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer in his commercial thermometers.

3.3 Design. The thermometers shall consist of a bimetallic heat sensitive element, dial, dial case and a window. The thermometers shall have a calibrated scale using the Fahrenheit (F) system or a combination Fahrenheit and Celsius (C) system and shall conform to the requirements of table I. The type I thermometer shall be accurate to ± 2° at 150° and 180° F. The type II thermometer shall be accurate within the following limits.

± 3° at 240°F
± 5° at 320°F
± 8° at 390°F

The type III thermometer shall be accurate to ± 25°F (see 4.3.1).

TABLE I. Graduations and ranges, °F

Type	Range minimum	Graduation maximum
I	130 to 200	10
II	200 to 400	2
III	100 to 600	25

3.4 Construction. The cases of the thermometers shall be fabricated of corrosion-resisting steel conforming to either ASTM A 167 or A 176. The thermometers shall be watertight (see 4.3.2).

3.4.1 Stems. The stems of type I and type II thermometers shall be not less than 1/4-inch in diameter. The length of the stem for type I thermometer shall be a minimum of 3-/78 inches and terminate in a point. The length of stem for type II thermometer shall be a minimum of 5-5/8 inches and terminate in a blunt end. The stems shall be attached to the dial face.

3.4.2 Case. The type III thermometer shall be designed to hang from and to stand on bar type oven racks (bars either parallel to or at right angles to door). The mounting shall be inclined backwards, 15° to 30° from the vertical.

3.4.3 Dial face. The dial face shall have a diameter of 2 inches, ± 1/8 inch, and shall be not less than 0.010 inch thick. The material used in the construction of the dial face and pointer shall be treated to resist corrosion. The dial face shall have a scale printed in contrasting color and shall be provided with manufacturer's standard pointer to indicate temperature. The minimum length of the outer edge of the printed scale shall measure 1-1/4 inches on the type I thermometer and 4-7/8 inches on the type II thermometer. The face on the thermometers shall indicate the Fahrenheit temperatures and terminology as follows:

Type I

180 - Turkey, Fresh
175 - Lamb, Medium
170 - Beef, Well Done
- Pork, Fresh
- Turkey, Roll
- Veal, Boneless
160 - Beef, Medium
- Ham, Cured
140 - Beef, Rare; Ham, Fully Cooked

Type II

320 to 338 - Caramel
300 to 309 - Hard Crack
270 to 290 - Soft Crack
250 to 266 - Hard Ball
240 to 250 - Firm Ball
235 to 239 - Soft Ball
219 to 223 - Jelly

Type III

100 to 250 - Warm
250 to 350 - Slow
350 to 400 - Moderate
400 to 450 - Hot
450 to 600 - Very Hot

3.4.4 Indicator. The type I and type II thermometers shall be equipped with a preset indicator, mounted on the bezel, capable of being moved to indicate any temperature on the dial.

3.4.5 Mounting clip. Type II thermometers shall be supplied with an adjustable spring clip which will hold the thermometer on a cooking kettle to keep the tip of the stem away from the side of the kettle.

3.5 Marking. Thermometers shall be marked in accordance with the manufacturer's commercial practice.

3.6 Finish. The thermometers shall have the manufacturer's standard commercial finish.

3.7 Workmanship. The finished thermometers shall be free from burrs, pits, stains, dirt, or grease. Numbering and graduation marks shall be clear and shall not be ragged or irregular.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Inspection of the end item. The inspection lot shall consist of all thermometers of one type offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated thermometer.

4.2.2.1 Visual examination. The end item shall be examine for defects in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Construction	Not type specified.	X	
Stem - (type I)	Does not terminate in a point.	X	
- (type II)	Does not terminate in a blunt end.		X
Dial	Not graduated as specified.	X	
	Not range specified.	X	
	Graduation marks ragged or irregular.	X	
	Scale not as specified.	X	
	Pointer missing.	X	
	Cooking temperature terminology or sugar stage terminology not as specified.		X
Indicators (type I and type II)	Missing or not preset type.	X	
Mounting clip - (type II)	Missing.	X	
	Not designed as specified.	X	
	Does not hold thermometer securely.	X	
Case (type III)	Not capable of standing on and hanging from oven racks.	X	
	Mounting not inclined backwards approximately 15° to 30° from vertical.		X
Finish	Burr or projection.	X	
	Not finished as specified.		X
	Tit, stain, or grease on the face or stem.	X	
Marking	Missing, incomplete, illegible, not as specified.		X

4.2.2.2 Dimensional examination. Dimensional examination shall be made of the thermometer to determine conformance to dimensions specified herein. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. Testing shall be performed on the end item in accordance with 4.3.1. Type I and type II thermometers shall be tested as specified in 4.3.2. The sample unit shall be one completely fabricated thermometer. The inspection level shall be S-4 with an AQL of 1.0 expressed in terms of defects per hundred units.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking are in conformance with section 5. Defects shall be scored as specified in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

TABLE III. Examination of preparation for delivery

Examination	Defect
Marking (exterior and interior container)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing or damaged or not as specified.
Contents	Number per container is more or less than required.

4.2.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.3 Tests.

4.3.1 Accuracy test. A standard test thermometer with a suitable range and graduation marks shall be used for the test. The standard thermometer shall bear current accuracy certification traceable to the National Bureau of Standards. Noncompliance with 3.3.1 constitutes failure of this test.

f figure 1 shall be suitable for flush mounting on

4.3.1.1 Type I and type II thermometers. The stem of the thermometer shall be placed into a constant temperature bath and compared to the standard thermometer. Type I thermometers shall be tested at 150°F and 180°F. Type II thermometers shall be tested at 240°, 320° and 390°F.

4.3.1.2 Type III thermometers. The type III thermometer shall be placed in a controlled temperature oven. The temperature shall be controlled from 200°F to 600°F and the test thermometer shall be compared to the standard thermometer.

4.3.2 Leakage test. Each sample thermometer shall be immersed in water at a temperature of 160°, ± 5°F for at least 20 minutes after which it shall be removed and immediately immersed in water at a temperature of 70° ± 10° F for at least 10 minutes. Evidence of condensation or water within the thermometer shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. Each thermometer shall be packaged in a snug-fitting setup paperboard box conforming to type I, variety 1, class A, style 3 or 4 of PPP-B-676. Each thermometer shall be immobilized by the use of die-cut pads or inserts made of the same material as the box, or wrapped in cushioning material conforming to type II, class A of PPP-C-843, or type III, style 1 PPP-P-291. Box closure shall be in accordance with the appendix of PPP-B-

5.1.2 Level C. The thermometers shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Fifty thermometers of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of the container specification.

5.2.2 Level B. Fifty thermometers of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fabricated and closed in accordance with PPP-B-636.

5.2.3 Level C. Thermometers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I thermometer is intended for use in measuring the internal temperature of meats and poultry during roasting to indicate the proper degree to which it is cooked. Type II thermometer is intended for use in indicating the correct temperature in the preparation for jams, jellies, or candy and the correct temperature of deep-fat frying oils. The type III thermometer is intended to be used for indicating temperatures in various types of ovens.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

GG-T-353D

Custodians:

Army - GL
Navy - YD
Air Force - 82

Review activities:

Army - MD
Navy - MS
DSA - GS

User activities:

Navy - MC, CG

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

VA-DMS
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